

TPWS

Train Protection and Warning System

Revision Guide — RS522 Issue 3 (December 2015)

⚠ All information sourced exclusively from RS522 Issue 3 ⚠

1. Purpose and Overview (§2.1)

1.1 What is TPWS?

TPWS stands for Train Protection and Warning System. It is designed to stop a train automatically if it passes a signal at danger or approaches certain signals or obstructions too fast.

KEY RULE: TPWS is NOT designed to prevent a Signal Passed at Danger (SPAD). Its purpose is to mitigate the consequences of a SPAD, an approach to a signal at danger at excessive speed, an approach to a speed reduction at excessive speed, or an approach to buffer stops at excessive speed.

1.2 Driver Responsibility

TPWS does NOT relieve the driver of their responsibility to observe and obey signals. The driver must continue to operate in accordance with the rules at all times.

1.3 Where TPWS is Provided (§2.1.3)

TPWS is provided at the following locations:

- Passenger line stop signals that protect crossing or converging movements.
- Non-passenger line stop signals that protect movements over passenger lines.
- Stop signals where an overlap would conflict with another signal's overlap or running line.
- Buffer stops — generally positioned approximately 65 m in advance, triggered at speeds greater than 10 mph.
- Permissive speed reductions — where the approach speed is 60 mph or more AND the reduction is at least one-third of that speed.

2. Track Equipment (§2.2)

2.1 Overview

TPWS track equipment consists of two systems: the Train Stop System (TSS) and the Overspeed Sensor System (OSS). The design of the equipment takes into account: braking performance of trains using the line, attainable speed of trains, distance to the conflict point, and gradient.

2.2 Train Stop System (TSS) — §2.2.2

The TSS consists of a loop mounted in the four-foot (between the rails) at the signal.

- When the signal is at DANGER — the TSS loop is energised.
- When the signal is at a PROCEED aspect — the TSS loop is de-energised.

When an on-train TPWS receiver passes over an energised TSS loop, a brake demand is immediately initiated.

TSS = one loop at the signal. Energised = danger. De-energised = proceed.

2.3 Overspeed Sensor System (OSS) — §2.2.3

The OSS uses two loops: an arming loop and a trigger loop. A timer starts when the train passes over the arming loop. If the train then passes over the trigger loop before the timer has expired, a brake demand is initiated. The set speed is determined by the distance between the arming and trigger loops.

- Freight train timers are calibrated 20% lower than passenger train timers (accounting for slower acceleration).
- OSS loops at stop signals are only energised when the signal is at DANGER.
- OSS loops at speed reductions and buffer stops are ALWAYS energised.

OSS = two loops (arming + trigger). If trigger passed before timer expires – brake demand. Freight timer is 20% lower.

2.4 Fitment (§2.2.4)

A signal fitted with TPWS will always have a TSS. It may also have one or more OSS fitted. Some signals are fitted with more than one OSS.

3. On-Train Equipment (§2.3)

3.1 Components (§2.3.1)

On-train TPWS equipment consists of:

- A receiver (aerial) at the front of the train.
- A TPWS control panel (standard or enhanced version).
- An AWS/TPWS acknowledgement button.
- A TPWS temporary isolation switch.
- An AWS/TPWS full isolation switch.

3.2 Receiver (§2.3.2)

The TPWS receiver is mounted at the front of the train. It senses when the train passes over TSS or OSS loops and sends signals to the on-train TPWS equipment.

3.3 Standard TPWS Control Panel (§2.3.3)

The standard panel has three controls/indicators:

Indicator / Button	Description
Brake Demand Indicator (RED)	Off = no brake demand. Flashing = brake demand initiated, awaiting acknowledgement. Steady = brake demand acknowledged.
Temporary Isolation / Fault Indicator (YELLOW)	Off = TPWS normal. Flashing = fault detected or start-up test failed. Steady = TPWS temporarily isolated.
Train Stop Override Button (YELLOW)	Used when authority has been given to pass a signal at danger. Overrides the TSS brake demand for approx. 20 seconds (passenger) or 60 seconds (freight). Illuminates when pressed; extinguishes

	when train passes over the TSS.
AWS/TPWS Acknowledgement Button	Used to acknowledge TPWS brake demands on the STANDARD panel. Press once flashing to change to steady. Brakes release 60 seconds after brake demand initiated.

IMPORTANT: There is NO audible warning when a TPWS brake demand is initiated (standard panel). There is a potential for confusion if the acknowledgement button is pressed 60 seconds or more after the initial brake demand — the indicator will immediately extinguish.

Self-test: the on-train TPWS carries out a self-test whenever the driving cab is opened. All indicators illuminate and TPWS applies the brakes. On successful completion, indicators extinguish and brakes release. If the test cannot be completed (e.g. receiver over an active loop), the Temporary Isolation/Fault indicator will flash.

3.4 Enhanced TPWS Control Panel (§2.3.4)

Some driving cabs are fitted with an enhanced TPWS control panel. It has three separate illuminated brake demand indicator buttons and a dedicated brake release button.

Indicator / Button	Colour	States & Description
SPAD indicator button	RED	Off / Flashing / Steady. Flashes with spoken message: "SPAD alert, contact signaller".
OVERSPEED indicator button	YELLOW	Off / Flashing / Steady. Flashes with spoken message: "Overspeed, contact signaller".
AWS indicator button	YELLOW	Off / Flashing / Steady. Used for AWS warnings only.
Brake Release button	—	Press simultaneously with the appropriate brake demand button to release brakes. If brake demand initiated by both overspeed AND AWS, pressing overspeed + brake release also clears the AWS-initiated brake demand.
Train Stop Override button	YELLOW	Same function as standard panel (20s passenger / 60s freight).

CRITICAL: On the ENHANCED panel, the AWS/TPWS acknowledgement button is used SOLELY for AWS warnings. It has NO involvement with TPWS. The brake release button is used with the appropriate brake demand button to release the brakes.

The enhanced panel also carries out a self-test when the cab is opened. All three indicators illuminate and TPWS applies the brakes. If the test is not successful, the Temporary Isolation/Fault

indicator flashes and the brakes remain applied. The on-train TPWS also carries out periodic self-tests in service; if a fault is detected the indicator flashes but brakes are NOT applied automatically.

4. Isolation Switches (§2.3.5 & §2.3.6)

4.1 Temporary Isolation Switch (§2.3.5)

The TPWS temporary isolation switch is mounted in the driver's cab but out of reach from the normal driving position.

- Used for certain operational movements that would otherwise cause an unintentional brake demand.
- Permitted uses include: movements during temporary block working, propelling movements, and movements within T3 possessions.
- The rules and regulations detail when the switch may be operated.

Temporarily isolating TPWS does NOT affect the AWS.

4.2 Full Isolation Switch (§2.3.6)

Traction units are fitted with AWS and TPWS full isolation switches. These may be separate or combined switches. They CANNOT be reached from the driving position. Some traction units have a visual indicator to advise the driver that the on-board AWS equipment is isolated.

- Used when there is a fault with the AWS system.
- If a train stops directly over an AWS magnet and the driver cannot cancel the AWS (and cannot change ends), it may be necessary to use the full isolation switch.

On cabs with a STANDARD TPWS panel: isolating the AWS also automatically isolates the TPWS.

5. Signal Box Equipment (§2.4)

5.1 Power Signal Boxes (§2.4.1)

In modern signalling installations, TPWS track equipment faults are indicated on the signalling panel or VDU workstation by a blue "fault" indication.

- On other panels, a TPWS fault at a signal is shown via a simulated "lamp out" fault in the signal lamp indication repeater circuit.
- During a TPWS track equipment failure, the signal indication on the panel will appear blank, even though the signal is displaying a red aspect.
- The signaller may not initially know whether it is the TPWS equipment or the signal lamp that has failed, and may ask the driver to confirm whether the signal is lit.

- In colour light areas, if there is a TPWS fault at a signal, the signal in rear will usually be held at danger until the affected signal displays a proceed aspect and its TPWS is no longer required to be energised.

5.2 Mechanical Signal Boxes (§2.4.2)

In mechanical signal boxes, the position of the lever in the frame determines the operational condition of the TPWS at that signal. Lever in normal position = signal at danger = TPWS armed.

A Failure Indication Unit (FIU) is provided to monitor the status of the TPWS (it is considered too restrictive to hold the signal in rear at danger for a TPWS fault in mechanical signalling).

FIU Event	What Happens
TPWS failure at a signal	Audible alarm sounds AND blue light flashes.
After alarm is acknowledged/cancelled	Blue light remains on (no longer flashes) as a reminder.
FIU test frequency	At least every 12 hours — at each shift change and when the signal box opens.

It is important that faults and failures are reported fully and promptly. This prevents important performance data from being lost, allows the signaller to advise subsequent drivers of defective equipment, and enables defective on-train equipment to be investigated without delay.

6. TPWS Fault Codes (§3 — Failures and Irregularities)

The following fault codes are specific to TPWS. They are used by drivers to report faults and irregularities to the signaller. (Codes 1–15 relate to AWS; codes 16 and 17 relate to TPWS.)

Fault Code	Fault Description
16	TPWS failed to activate (when it should have activated).
17	TPWS operated when not required (activated when it should not have).

Prompting reporting of wrong-side failures (e.g. TPWS activated when not required — code 17) is essential so that defective equipment can be investigated without delay.

7. Key Comparisons: Standard vs Enhanced Panel

Feature	Standard Panel	Enhanced Panel
Brake demand indicator	Single red light (one indicator for all brake demands)	Three separate buttons: SPAD (red), OVERSPEED (yellow), AWS (yellow)
Audible / spoken alert	No audible warning on TPWS brake demand	Spoken message with alert tone: "SPAD alert, contact signaller" or "Overspeed, contact signaller"
Releasing brakes	Brakes release automatically 60	Press brake release button AT

	seconds after brake demand. AWS/TPWS ack button changes indicator from flashing to steady.	THE SAME TIME as the appropriate brake demand indicator button.
AWS/TPWS ack button role	Acknowledges BOTH AWS warnings AND TPWS brake demands	Used SOLELY for AWS warnings — NO involvement with TPWS
Self-test on cab opening	Indicators illuminate + brakes apply. Success = extinguish.	Same as standard PLUS: if fault found, brakes remain applied.
Periodic self-test (in service)	Not described separately for standard	If fault detected: Temp Isolation/Fault indicator flashes but brakes NOT applied automatically

Feature	TSS	OSS
Number of loops	One loop	Two loops (arming + trigger)
Position	In the four-foot at the signal	In the four-foot, spaced apart
How it works	Receiver passes over energised loop → immediate brake demand	Timer starts at arming loop; if trigger loop passed before timer expires → brake demand
When energised (at stop signals)	Energised when signal at DANGER; de-energised at PROCEED	Energised when signal at DANGER; de-energised at PROCEED
When energised (speed reductions / buffer stops)	N/A — TSS is only at signals	ALWAYS energised
Freight timer	N/A	20% lower than passenger timer

8. Questions and Answers

Q1. What does TPWS stand for?

A1. Train Protection and Warning System.

Q2. What four situations is TPWS designed to stop a train in?

A2. (1) Passing a signal at danger (SPAD). (2) Approaching a signal at danger at excessive speed. (3) Approaching a speed reduction at excessive speed. (4) Approaching buffer stops at excessive speed.

Q3. Does TPWS relieve the driver of their responsibility to obey signals?

A3. No. TPWS does NOT relieve the driver of any responsibility. Drivers must continue to observe and obey signals at all times.

Q4. Is TPWS designed to prevent SPADs?

A4. No. TPWS is NOT designed to prevent SPADs. It is designed to mitigate the consequences of a SPAD.

Q5. At what approximate distance ahead of buffer stops is TPWS fitted, and at what speed does it activate?

A5. Approximately 65 m in advance of the buffer stops. It activates at speeds greater than 10 mph.

Q6. What two conditions must be met for TPWS to be provided at a permissive speed reduction?

A6. (1) The approach speed must be 60 mph or more. (2) The speed reduction must be at least one-third of the approach speed.

Q7. What is the TSS and where is it fitted?

A7. The Train Stop System (TSS) is a single loop mounted in the four-foot (between the rails) at the signal. It is energised when the signal is at danger and de-energised when the signal shows a proceed aspect.

Q8. How does the OSS work?

A8. The OSS (Overspeed Sensor System) has two loops: an arming loop and a trigger loop. A timer starts when the train passes over the arming loop. If the train then passes over the trigger loop before the timer expires, a brake demand is initiated.

Q9. How is the OSS set speed determined?

A9. The set speed is determined by the distance between the arming loop and the trigger loop.

Q10. How does the freight train OSS timer differ from the passenger timer?

A10. The freight timer is calibrated 20% lower than the passenger timer, to account for the slower acceleration of freight trains.

Q11. When is the OSS at a stop signal energised?

A11. Only when the signal is at danger. It is de-energised when the signal shows a proceed aspect.

Q12. When is the OSS at speed reductions and buffer stops energised?

A12. Always — it is permanently energised at speed reductions and buffer stops.

Q13. Name the five components of on-train TPWS equipment.

A13. (1) A receiver/aerial at the front of the train. (2) A TPWS control panel (standard or enhanced). (3) An AWS/TPWS acknowledgement button. (4) A TPWS temporary isolation switch. (5) An AWS/TPWS full isolation switch.

Q14. Describe the three states of the brake demand indicator on a standard TPWS panel.

A14. Off = no brake demand. Flashing = brake demand initiated, awaiting acknowledgement. Steady (on) = brake demand has been acknowledged.

Q15. On a standard TPWS panel, is there an audible warning when a TPWS brake demand is initiated?

A15. No. There is no audible warning on a TPWS brake demand on the standard panel.

Q16. How long after the brake demand was initiated will the brakes release on a standard panel?

A16. 60 seconds after the brake demand was initiated.

Q17. What is the potential for confusion if the acknowledgement button is pressed 60 or more seconds after the initial brake demand?

A17. If the button is pressed 60 or more seconds after the initial brake demand, the indicator will immediately extinguish — potentially making it unclear to the driver what caused the brake demand.

Q18. What does the yellow Temporary Isolation/Fault indicator show when it is flashing on a standard panel?

A18. A fault has been detected in the on-train TPWS equipment, or the start-up test has not been completed successfully.

Q19. What does the yellow Temporary Isolation/Fault indicator show when it is on steady on a standard panel?

A19. The train TPWS equipment has been temporarily isolated.

Q20. What happens during the TPWS self-test when a driving cab is opened (standard panel)?

A20. All indicators illuminate and TPWS applies the brakes. If the test completes successfully, the indicators extinguish and the brakes release. If unsuccessful (e.g. receiver over an active loop), the Temporary Isolation/Fault indicator flashes.

Q21. Name the three brake demand indicator buttons on an enhanced TPWS panel and their colours.

A21. SPAD — red. OVERSPEED — yellow. AWS — yellow.

Q22. What spoken messages can be announced on an enhanced TPWS panel?

A22. 'SPAD alert, contact signaller' or 'Overspeed, contact signaller'. Each is preceded by an alert tone and repeated until the brake application is acknowledged.

Q23. How are the brakes released on an enhanced TPWS panel?

A23. By pressing the brake release button at the same time as pressing the appropriate brake demand indicator button.

Q24. On an enhanced panel, if a brake demand was initiated by both an overspeed and by AWS, how is it cleared?

A24. Pressing the OVERSPEED brake demand button together with the brake release button also releases the AWS-initiated brake demand.

Q25. What is the role of the AWS/TPWS acknowledgement button on an enhanced TPWS panel?

A25. On the enhanced panel it is used SOLELY to acknowledge AWS warnings. It has NO involvement with TPWS.

Q26. What is the Train Stop Override button used for?

A26. It is used — where authority has been given — to override the brake demand from a TSS loop when passing a signal at danger. It is effective for approximately 20 seconds (passenger) or 60 seconds (freight).

Q27. When does the Train Stop Override button extinguish?

A27. It extinguishes when the train passes over the TSS loop.

Q28. What is the TPWS temporary isolation switch used for?

A28. For operational movements that would otherwise cause an unintentional brake demand — such as temporary block working, propelling movements, or movements within T3 possessions.

Q29. Does temporarily isolating TPWS affect the AWS?

A29. No. Temporarily isolating TPWS does NOT affect the AWS.

Q30. What is the full isolation switch used for?

A30. It is used when there is a fault with the AWS system. It cannot be reached from the driving position.

Q31. On a cab fitted with a standard TPWS panel, what happens when the driver isolates the AWS?

A31. The TPWS will be isolated automatically.

Q32. How is a TPWS track equipment fault indicated in a power signal box?

A32. In recent installations by a blue 'fault' indication on the panel or VDU workstation. On other panels, by a simulated 'lamp out' fault in the signal lamp indication repeater circuit.

Q33. During a TPWS track equipment failure, what will the signal indication on a power signal box panel show?

A33. The panel will appear blank (as if the signal lamp has failed), even though the signal is displaying a red aspect.

Q34. In a colour light area with a TPWS fault, what usually happens to the signal in rear?

A34. The signal in rear is usually held at danger until the affected signal displays a proceed aspect and its TPWS is no longer required to be energised.

Q35. What equipment is provided in a mechanical signal box to monitor TPWS status, and why?

A35. A Failure Indication Unit (FIU). It is provided because it is considered too restrictive to hold the signal in rear at danger for a TPWS fault in mechanical signalling.

Q36. What does the FIU do when a TPWS failure occurs at a signal?

A36. An audible alarm sounds and a blue light flashes. Once the alarm is acknowledged and cancelled, the blue light remains on (steady) as a reminder.

Q37. How often must the FIU test be performed?

A37. At least every 12 hours — at each shift change and when the signal box opens.

Q38. What are the two TPWS-specific fault codes and what do they mean?

A38. Code 16: TPWS failed to activate (when it should have). Code 17: TPWS operated when not required (activated when it should not have).

Q39. Why is it important to report TPWS faults fully and promptly?

A39. It prevents important performance data from being lost, allows the signaller to advise subsequent drivers of defective equipment, and enables defective on-train equipment to be investigated without delay.

Q40. A TPWS brake demand is initiated. TRUE or FALSE: On a standard panel, there will be an audible alarm to alert the driver.

A40. FALSE. There is no audible warning on a TPWS brake demand on the standard TPWS control panel.

9. Defective On-Train TPWS Equipment (GERT8000-TW5)

The following rules apply when TPWS on-train equipment is defective. Source: GERT8000-TW5 Issue 14.

9.1 Starting a Journey from a Maintenance Depot (\$25.1)

You must NOT allow a train to start a journey if the TPWS is defective in any cab which is required to be driven from when TPWS is required to be in operation.

9.2 Starting a Journey from Somewhere Other than a Maintenance Depot (\$25.2)

You CAN allow a train to start a journey with TPWS defective in the cab to be driven from, as long as TPWS will NOT be required to be in operation during the journey. You must:

- Tell the train operator's control at the first convenient opportunity.
- Carry out any instructions given.

You CAN allow a train to start a journey (but NOT carrying passengers) with TPWS defective when TPWS is required to be in operation, in order to travel to a maintenance depot for repair, as long as you:

- Tell the signaller.
- Get permission for the train to start a journey in this condition.

9.3 During a Journey (\$25.3)

If TPWS becomes defective when it SHOULD be in operation:

- Stop your train immediately.
- Tell the signaller.
- Do not move the train until instructed to do so.
- Carry out the instructions given.

If TPWS becomes defective when it is NOT required to be in operation:

- Tell the train operator's control at the first convenient opportunity.
- Carry out any instructions given.

9.4 Failure to Activate (\$25.4)

If you become aware that TPWS has FAILED TO ACTIVATE when it should have done:

- Stop your train immediately.
- Tell the signaller.
- Do not move the train until instructed to do so.
- Carry out the instructions given.

Failure to activate = TPWS fault code 16. Stop immediately and tell the signaller.

9.5 Conditions when Permitted to Proceed with Defective TPWS (\$25.5)

If permission is given to start a journey or proceed after TPWS has become defective, you must follow these conditions during any part of the journey where TPWS would normally be in operation:

Situation	Driver Action
No competent person provided	Freight trains and OTMs: not exceeding 50 mph (80 km/h). All other trains: not exceeding 60 mph (95 km/h), or any lower permissible speed that applies — until a competent person is provided.
Competent person is provided	Proceed at normal permissible speed.

9.6 Competent Person Travelling with Driver (\$2.2)

If AWS or TPWS fails, a competent person may be provided to travel with the driver. When accompanied by a competent person, the driver must tell them which equipment is defective and what to do.

When approaching a signal, the DRIVER must:

- Call out the signal aspect or indications to the competent person.
- Give a commentary on speed reduction on the approach to cautionary and stop aspects.
- Tell the competent person when applying brakes for speed restrictions.

The COMPETENT PERSON must:

- Acknowledge the driver's reaction to signal aspects, sequences or indications.
- If necessary, remind the driver of a signal displaying a cautionary or stop aspect.
- Acknowledge the driver's reaction to speed restrictions.
- If necessary, remind the driver of a speed restriction ahead.

SUMMARY: Defective TPWS in operation – STOP IMMEDIATELY – tell signaller – await instructions. Not in operation – tell control – await instructions.

10. Additional Questions and Answers — Defective TPWS (TW5)

Q41. Can a train start a journey from a maintenance depot if TPWS is defective in the cab to be driven from and TPWS is required to be in operation?

A41. No. You must NOT allow a train to start a journey in these circumstances.

Q42. A train is starting from somewhere other than a maintenance depot. TPWS is defective but will NOT be required to be in operation during the journey. What must the driver do?

A42. The driver may start the journey but must tell the train operator's control at the first convenient opportunity and carry out any instructions given.

Q43. Can a train with defective TPWS start a journey carrying passengers to travel to a maintenance depot for repair?

A43. No. The train must NOT be carrying passengers. Permission can only be given for an empty working to travel to a maintenance depot.

Q44. TPWS becomes defective during a journey when it should be in operation. What must the driver do?

A44. Stop the train immediately. Tell the signaller. Not move the train until instructed to do so. Carry out the instructions given.

Q45. TPWS becomes defective during a journey when it is NOT required to be in operation. What must the driver do?

A45. Tell the train operator's control at the first convenient opportunity and carry out any instructions given. There is no requirement to stop immediately.

Q46. TPWS has failed to activate when it should have. What must the driver do?

A46. Stop the train immediately. Tell the signaller. Not move the train until instructed to do so. Carry out the instructions given.

Q47. Permission is given for a passenger train to proceed with defective TPWS and no competent person is provided. What is the maximum speed?

A47. Not exceeding 60 mph (95 km/h), or any lower permissible speed that applies.

Q48. Permission is given for a freight train to proceed with defective TPWS and no competent person is provided. What is the maximum speed?

A48. Not exceeding 50 mph (80 km/h), or any lower permissible speed that applies.

Q49. If a competent person IS provided, what speed restriction applies to a train with defective TPWS?

A49. Normal permissible speed — no additional speed restriction applies.

Q50. On a TCB line, what must the signaller ensure when allowing a train with defective TPWS to proceed?

A50. At least two controlled signals must be kept at danger between the train with defective TPWS and any conflicting or converging movements ahead of it.

Q51. On an absolute block line, when can the signaller accept a train with defective TPWS?

A51. Only when the line is clear to the section signal. If the home signal is also the section signal, the train must first have been accepted by the next signal box.

Q52. What must the driver call out when accompanied by a competent person due to defective TPWS, on the approach to a signal?

A52. The signal aspect or indications, and a commentary on the speed reduction on the approach to cautionary and stop aspects.

Q53. What is the competent person's role when accompanying a driver with defective TPWS?

A53. To acknowledge the driver's reaction to signal aspects and speed restrictions, and if necessary to remind the driver of cautionary/stop aspects or speed restrictions ahead.

Sources: RS522 Issue 3 (AWS and TPWS Handbook) — December 2015 | GERT8000-TW5 Issue 14 — Defective On-Train Equipment or Isolated Vehicles